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ADVANCED REVIEW

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Y-chromosome short tandem repeats in forensics—Sexing, profiling, and matching male DNA

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Abstract

The analysis of short tandem repeat (STR) markers located on the Y chromosome is an established method in forensic casework analysis. Usually this method is applied in cases of male-on-female sexual assault, in which the victim's DNA is in great excess and masked the male contribution. Y-STR analysis is able to detect the presence of minuscule amounts of male DNA of one or multiple donors and resulting genetic profiles can be compared to known reference samples. The expert has to determine whether the Y-STR result of a trace is suitable for a biostatistical calculation in case of a match. The Discrete Laplace method which is implemented in the Y chromosome haplotype reference database (YHRD) can be used to estimate haplotype frequencies with a better approximation than other methods namely the counting method. Moreover, Y-SNPs in combination with Y-STRs can infer the biogeographical origin of an unknown male person with comparably high precision due to the availability of a robust phylogenetic tree and large reference data collections. Y-based ancestry prediction and familial searching can, therefore, provide important investigative leads in crime cases without suspect.

This article is categorized under:

Forensic Biology > Haploid Markers
Forensic Biology > Ancestry Determination using DNA Methods
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KEYWORDS

ancestry inference, haplotype, Y chromosome, YHRD, Y-STR

1 | INTRODUCTION

Short tandem repeats (STRs) located on the Y chromosome have been used in forensic investigations since 1990 when the first case was reported in Germany. In this highly publicized case a woman was murdered after having sexual intercourse. A suspect was apprehended. The victim's blood and the vaginal swab containing sperm cells were collected. The sperm were separated from female cells by differential lysis. Conventional hybridization techniques using DNA probes failed to work in the trace, and PCR analysis of four recently developed STRs was used instead, two located on chromosome 12, one on

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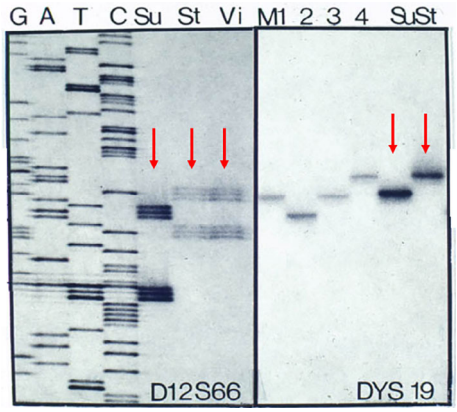


FIGURE 1 Polyacrylamide vertical slab gel using a sequenced DNA as length standard (four lanes to the left). Autosomal length alleles at STR locus D12S66 and Y-STR alleles at locus DYS19 were detected in a vaginal swab (St) and reference DNA samples of suspect (Su) and victim (Vi). Alleles are marked by arrows. M1-4 depict DYS19 reference alleles

chromosome 6 and one on chromosome Y. The Y chromosomal locus DYS19 could not be amplified in the nonsperm fraction of the trace, but generated alleles of different lengths for the sperm fraction and the suspect's blood. Based on this unequivocal result from the Y chromosome the suspect was excluded as the donor of the stain and released from jail (Roewer & Epplen, 1992) (Figure 1).

This first case proves that Y chromosome STR analysis is an excellent method to verify the presence of male DNA in mixed stains and to exclude a reference person as the donor of the male cells. Forensic experts around the world recognized the potential of the method for sexing and exclusion. "Y-chromosome analysis urged for sex crimes" the journal *Nature* headlined (Schiermeier, 2001). Only recently, however, the Sexual Assault Forensic Evidence Reporting (SAFER) Working Group empaneled by the National Institute of Justice (USA) published a document on national best practices for sexual assault kits (SAK), which recommends a high-throughput process flow with primary (database up loadable autosomal STRs) and secondary testing (Y-STRs) (NIJ, 2017). This binary workflow has been validated before (Hanson & Ballantyne, 2014; Hanson & Ballantyne, 2016; Purps, Geppert, Nagy, & Roewer, 2015). Continuously discussed, however, are questions on the correct and fair interpretation, namely on the value of a match between two Y chromosomal profiles that are inherited along the paternal lineage and carrying mutations over many generations.

A case from Germany occurring in 2015 could, therefore, mark the beginning of a new epoch in Y-STR testing. In an unsuccessful attempt of robbery a woman was murdered in her apartment. A suspect was apprehended. A tape used to gag the woman contained on a background of massive female saliva some epithelia from a male person. Due to the well-known effect of preferential amplification conventional autosomal analysis failed to detect the male contribution, but Y-STR analysis generates a full profile for 23 loci. The profile matches the suspect who was arrested. Furthermore, the paternal genealogy of the suspect was investigated and seven male relatives in three generations were typed for not less than 47 Y-STRs. All paternal relatives who are connected by

Minimal | PowerPlex Y | Yfiler | PowerPlex Y23 | Yfiler Plus | Maximal

Report for Sample #1

Sample Name: Manual input

Metapopulation*	Observations[1] in YHRD	Augmented Counting[1] (n+1/N+1)	Discrete Laplace (DL)[1]	Proportion (Western Europe = 1)
Western European	0	1 / 26,386	1 / 49,715	
Eastern European	0	1 / 8,379	1 / 1,750,172	1 : 35
Semitic[2]	0	1 / 5,188	1 / 1,262,344	1 : 25
Altaic[3]	0	1 / 10,055	1 / 2,098,588	1 : 42
Indian	0	1 / 6,028	1 / 12,339,871	1 : 248
Subsaharan African	0	1 / 2,347	1 / 22,075,700	1 : 444
Chinese (Han)	0	1 / 54,395	1 / 236,007,081	1 : 4,747

[1] retrieved from https://yhrd.org (release 59)
[2] includes Arab speaking populations
[3] includes Turk speaking populations

FIGURE 2 YHRD report for a 17-loci Y-STR profile extracted from a female/male mixed DNA trace described in the text. Observations, augmented counts (in the formula "n" depicts the observations and "N" the database size, the haplotype in question is added to both numerator and denominator resulting in a conservative "augmented" count estimate) and DL estimates are given for different metapopulations. The count and the extrapolated DL values are in the same order of magnitude only for Western Europeans (depicted in red color), but become clearly divergent for Asian and African reference databases which are dominated by different founder lineages. The analyzed haplotype is blinded

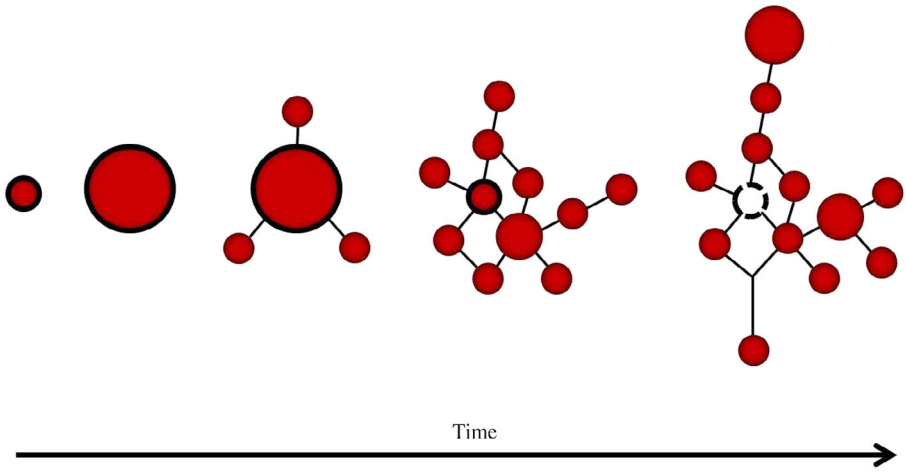


FIGURE 3 Schematic representation of a Y-STR cluster and its evolution from an ancestral haplotype (modified from Jobling & Tyler-Smith, 2003)

11 germ line transfers match the suspect's 47-loci Y-STR profile and the 23 loci of the trace DNA. The case was brought to court and since the DNA results were considered decisive here, the match statistics became a central part of the trial. For the first time in Germany a model-based extrapolation method (Andersen, Eriksen, & Morling, 2013), which makes use of an in-depth analysis of Y-STR profiles in more than thousand worldwide population samples, was used in a court case. Given the evidence the expert argues that other men sharing the high-resolution profile of the suspect exist in the population and that these men obviously belong to one family. The chance that identical 47-loci haplotypes generated by random mutation would exist outside the extended family is very low as explained in the following. The judgment on the number of further unknown relatives plus possible unrelated males with profile identity by chance in the suspect population, in other words the "rarity" of the haplotype, was hitherto assessed based on observations in a randomly sampled database, a method widely known as "naïve" or "augmented counting" depending on whether the haplotype is added to the database or not. From Figure 2 it becomes obvious that the augmented count estimators are bounded from below by the inverse of the database sizes creating large error rates (Roewer, 2016). Preanalysis of the database and modeling the probability distribution of its haplotypes, however, can uncover the hidden relatedness structure within a reference population which consists of haplotypes organized in founder-derived descent clusters interconnected by mutational events (Figure 3).

Once this relatedness structure is elucidated, the proximity or distance of the target profile to the haplotypes in the network can be analyzed and values can be retrieved. From these values a probability of occurrence of the target haplotype in any representative population can be calculated. Using this procedure, called Discrete Laplace method (Andersen et al., 2013), a probability of close to 1 in 50,000 for the occurrence of the reduced Y-STR profile (17 loci) in the Western European metapopulation was conservatively calculated (Figure 2). The court followed the argumentation of the experts saying that the perpetrator most probably belongs to a certain family in Germany. Only the suspect but none of the other known relatives has a reason or the ability to commit the crime. The suspect was sentenced to life imprisonment in the first instance. The defense was unsatisfied with the outcome and appealed. Eventually the highest appellate court in Germany, the *Bundesgerichtshof (BGH)*, confirmed the judgment. The case strongly influences the process of drafting Y-STR interpretation guidelines in Germany. The document, which was finally released in April 2018 recommends the use of the Discrete Laplace method instead of counting and advised the use of representative metapopulations from the Y chromosome haplotype reference database (YHRD) database as proxies of suspect populations (Roewer & Willuweit, 2018; Willuweit, Anslinger, et al., 2018; Willuweit, Weirich, Schneider, & Roewer, 2018).

Twenty-five years passed between the two cases. Much time was needed to form the scientific foundation for male-specific DNA analysis. We have now a good understanding of the structural peculiarities of the Y chromosome and understand better the complex interplay of factors, which influence its diversity and distribution patterns. Just as much time and effort was spent to develop and validate robust analytical tools with the ability to detect minuscule male DNA amounts in mixed body fluids or contact stains. Finally, the implementation of the method into the high-throughput workflow, which is typical for modern crime labs, is laborious and requires extra personnel and training. Not to forget the work on a framework of guidelines and recommendations, which provide common ground for court-going experts.

Getting into details of this process, two major turning points in forensic Y chromosome analysis need to be emphasized—the availability of highly discriminating multiplex kits and the launch of online reference databases.

2 | Y-STR MARKERS

At the turn of the millennium the first multiplex PCR kit for routine application with 12 STR loci was developed and validated (Shewale et al., 2004). The panel includes the widely used "minimal haplotype" (Kayser et al., 1997) plus two additional markers (Ayub et al., 2000). In the following years, the number of markers increased steadily. Currently different kits and supplementary multiplexes can be combined to amplify more than 40 Y-STR sequences, a required number to pursue a strategy "typing until exclusion" in forensic casework (Table 1). All of these Y-STRs are carefully selected from a large reservoir of candidate sequences obtained from the male-specific region of the Y chromosome (Ballantyne et al., 2010; Redd et al., 2002). Detection of allelic variants is possible with capillary electrophoresis (CE) for length or massively parallel sequencing (MPS) for length and sequence variability in the repeat and the flanking sites (Just, Moreno, Smerick, & Irwin, 2017) (Table 1). MPS analysis of STR loci increased the number of variant alleles per locus and consequently the diversity (Kwon, Lee, Kim, Lee, & Shin, 2016). To facilitate the description of sequence-based alleles a catalog of sequence diversity at Y chromosomal STR loci has been published (Gettings et al., 2017). However, a recent validation study showed that the performance of MPS-based Y-STR analysis for the detection of the minor male component in female/male mixtures is currently not nearly as good as the CE method (Köcher et al., 2018).

3 | REFERENCE DATABASES

The second turning point was the establishment of online accessible population databases which consist of Y-STR profiles typed in reference populations (Roewer et al., 2001, Redd et al., 2006). The largest known database, the YHRD, now includes more than a hundred national databases which were populated in the past 20 years and continue to grow at a steady pace until today (Figure 4). The YHRD project started in 1999 with a collection of 3,825 haplotypes typed in 48 populations for up to 13 Y-STRs. Currently, the 59th edition of the YHRD is online with 265,324 profiles from 1,251 populations typed for up to 29 Y-STRs and a large number of SNPs (Roewer, 2016; Willuweit & Roewer, 2015). Resources include primary data such as regularly and irregularly spaced, null and multiple alleles, haplotypes in different formats, mutation rates and Y-SNP data. These data facilitate statistical analyses for a range of casework applications and interpretation purposes, as, for example, kinship and mixture analysis, estimation of haplotype centers for Discrete Laplace calculation, AMOVA including MDS visualization and ancestry prediction based on Y haplogroups. All incoming data are validated first by the submitter using an online tool (https://yhrd.org/pages/tools/validator) and then by the curators before being uploaded and accepted for publication. In the following, the population structure of the YHRD database is explained in more detail.

4 | YHRD MIRRORS THE WORLD'S Y CHROMOSOME DIVERSITY

This database is not like a police database filled by law enforcement agencies with DNA profiles connected to known offenders or crime scene traces. YHRD is a reference database with anonymous Y-STR profiles (20% of which rooted by Y-SNPs) from sampling projects guided by academic research. YHRD does not provide information on an individual match but instead uses bulk population data to calculate frequencies and other parameters in order to assess the evidential value of a Y-STR profile match. As the database grew, it became clear how inhomogeneously Y-STR diversity is distributed across the world. Using population genetic methods the differentiation within and between populations can be analyzed. One such method called analysis of molecular variance (AMOVA) (Excoffier, Smouse, & Quattro, 1992) revealed hitherto unknown gradients for Y-STR haplotypes reflecting historical and prehistoric demographic events and the effects of extended patrilineal relationship. The earliest AMOVA analysis compared two samples with less than a hundred participants each from regions in Western Netherlands and Eastern Germany. A small but significant genetic distance was measured for the haplotype pools (Roewer et al., 1996). This rather surprising result provoked larger studies for 91 European populations with more than 12,000 samples and here a bigger picture was rolled out (Roewer et al., 2005). In the center of Europe a clear division between an Eastern and a Western European Y chromosome pool was ascertained and thus explained the interpopulation variance between the Dutch and Germans. Whereas the Dutch region fits well into a Northwestern cluster, the population of East Germany received different haplotypes from the Romance-speaking and the Slavic-speaking parts of Europe thus creating a measurable genetic distance. An abundance of studies using Y-STRs and Y-SNPs confirm that the occupation of Europe by male migrants in the past creates different pools of Y chromosome lineages in the West, East, Southeast or the Northeast (Diaz-Lacava et al., 2011; Rosser et al., 2000). Meta-studies on a global scale (Kayser et al., 2001; Purps et al., 2014) revealed

TABLE 1 A list of frequently used Y-STR panels in forensic practice

Marker	Regular copy number[a]	17 marker (CE)[b]	23 marker (CE)	27 marker (CE)	23 marker (Seq)[c]	26 marker (Seq)	RM panel[d]
DYS19	1	*	*	*	*	*	
DYS385	2	*	*	*	*	*	
DYS389I	1	*	*	*	*	*	
DYS389II	1	*	*	*	*	*	
DYS390	1	*	*	*	*	*	
DYS391	1	*	*	*	*	*	
DYS392	1	*	*	*	*	*	
DYS393	1	*	*	*	*		
DYS437	1	*	*	*	*	*	
DYS438	1	*	*	*	*	*	
DYS439	1	*	*	*	*	*	
DYS448	1	*	*	*	*	*	
DYS449	1			*			*
DYS456	1	*	*	*	*		
DYS458	1	*	*	*	*		
DYS460	1			*		*	
DYS481	1		*	*	*	*	
DYS505	1					*	
DYS518	1			*			*
DYS522	1					*	
DYS526	1						*
DYS533	1		*	*	*	*	
DYS547	1						*
DYS549	1		*		*	*	
DYS570	1		*	*	*	*	*
DYS576	1		*	*	*	*	*
DYS612	1					*	*
DYS626	1						*
DYS627	1			*			*
DYS635	1	*	*	*	*	*	
DYS643	1		*		*	*	
GATA H4	1	*	*	*	*	*	
DYF387S1	2			*		*	*
DYF399S1	3						*
DYF403S1	3						*
DYF404S1	2						*
Sum: markers/copies	36/43	16/17	22/23	25/27	22/23	24/26	13/19

[a]Multicopy markers will be amplified at more than one locus.
[b]CE, capillary electrophoresis.
[c]Seq, massive parallel sequencing.
[d]Rapidly mutating marker panel (Alghafri, Goodwin, Ralf, Kayser, & Hadi, 2015).

further knowledge on the male-mediated expansions leading to the regional cluster structure of Y-STR haplotypes (Balaresque et al., 2015; Batini & Jobling, 2017). The term "metapopulation" was adapted to describe an assemblage of Y-STR profiles spread over a territory, which descend from one or several common ancestral lineages (Millstein, 2010; Willuweit & Roewer, 2015). Population genetic analyses on different Y chromosomal marker sets show that metapopulations are stabilized over time by cultural and social factors including a common language (Baker, Rotimi, & Shriner, 2017; Quintana-Murci et al., 2001), patrilocality (Oota, Settheetham-Ishida, Tiwawech, Ishida, & Stoneking, 2001) and/or geographical barriers (Rosser et al., 2000). Consequently, the YHRD was reorganized with the activation of YHRD version 4.0 in August 2014 in order to reflect Y-specific metapopulation structures. Haplotype frequencies can still be reported country-wise (135 political entities are searchable) but preferably in metapopulations (MP), as, for example, in a Western European MP or a Sino-Tibetan MP (Table 2).

Contributors to the YHRD are requested to provide metadata (geographic coordinates, ethnic ancestry and language group) on their population samples in order to assign these sensibly to an MP. The reporting guidelines in Germany from 2018 (Willuweit, Anslinger, et al., 2018, Willuweit, Weirich, 2018) explicitly recommend MPs as proxies for suspect populations. With an interactive upload and validation program the YHRD became a widely accepted online platform for peer-reviewed population studies. The current release 59 (from 2018/Nov/01) represents 1,251 population studies, which are described in more than 540 papers. According to a decision of Scientific Working Group on DNA Analysis Methods (SWGDAM) in 2018 the US-based Y-STR database was permanently transferred to the YHRD. Forensic science providers in countries like UK or Singapore created national databases and transferred it to the YHRD platform. This international cooperation is key for a fair interpretation of Y-STR results in a population-based context. The International Society of Forensic Genetics (ISFG) plans to publish guidelines for Y-STR interpretation by the end of 2019 in order to serve the wider forensic genetics community.

5 | FREQUENCY AND PROBABILITY CALCULATIONS FOR Y-STR PROFILES

Statistical methods relying on the independence of loci (as in the autosomal case) cannot be applied to Y chromosomal profiles. All Y-STRs, which constitute a haplotype are linked together and are not inherited independently but *en bloc* along

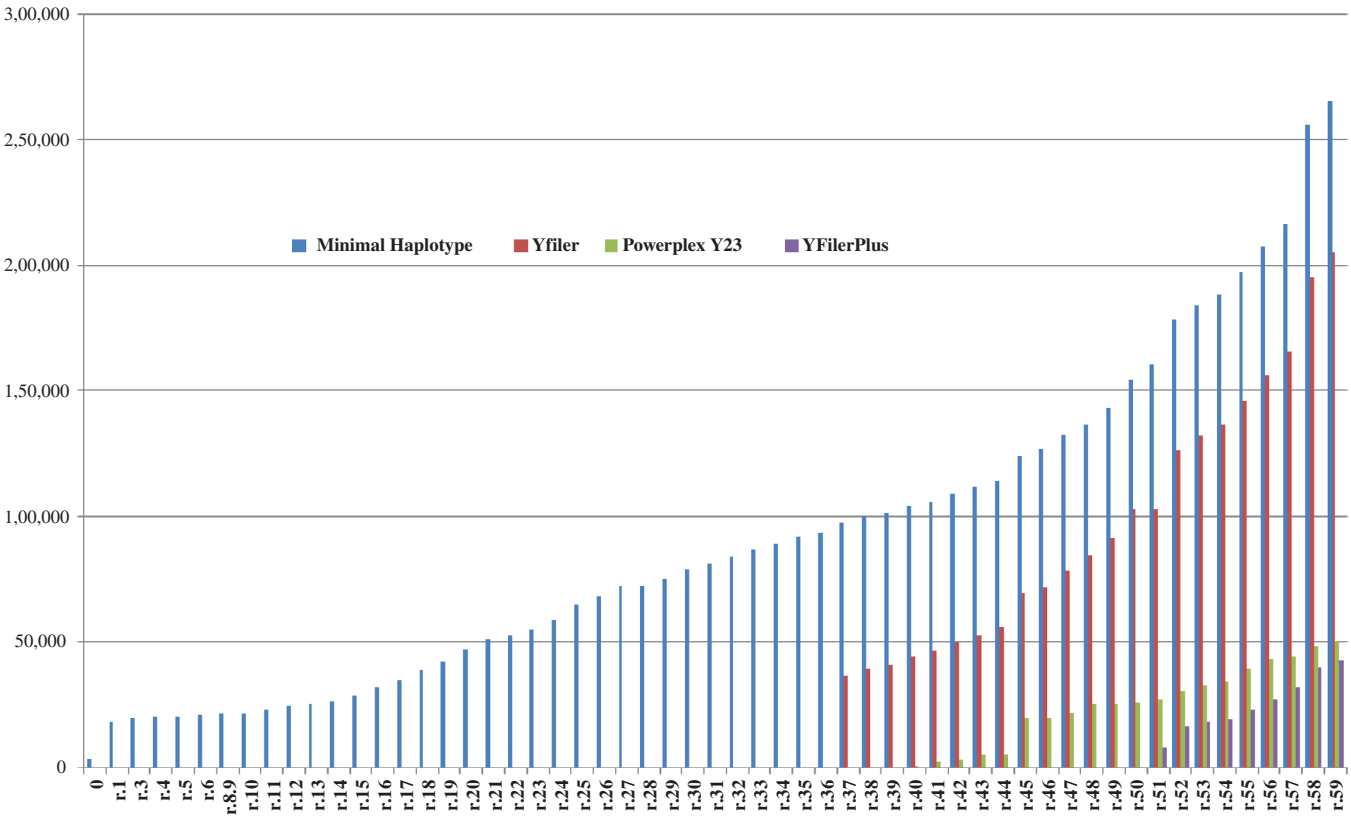


FIGURE 4 Growth of YHRD between 1999 to 2018 (Release 1 to 59)

paternal lineages. In a crime case it is, therefore, safe to assume that besides the true trace DNA donor other men with the identical profile exist. Clearly, close relatives can carry identical haplotypes, but also unrelated males may display the same haplotype as the true donor. The "relatives" proportion of identical haplotypes is called "identical by descent (IBD)", and the haplotypes, which arise by random mutation are "identical by state (IBS)" (de Knijff, 2000). The number of IBD and IBS haplotypes returned as matches by a database query is mainly influenced by the following factors: number of loci per haplotype, the mutation rates of these loci, the database size and the extent of family and clan structures in the sample population. Obviously, the higher the number of markers per haplotype is and the higher their mutation rate, the higher the probability that all matching haplotypes are IBD, and not anymore IBS. More complex is the issue of "random sampling" and here the social organization and family structures in populations or countries are of key interest to guide the sampling projects. With ethnographic data compiled in George Murdock's Ethnographic Atlas (Murdock, 1967) Emmanuel Todd provides world-scale historical classifications of family systems and relatedness patterns in different regions of the world. For example, mobile and independent nuclear families are typical for the Western world, while the community family type where the family is embedded in (mostly) patrilineal relationship systems with co-residence is still dominant in large parts of Eurasia (Todd, 2011). Haplotypes will, therefore, always reflect to some extent the prevalent relatedness system in the sample population. The already cited multicentre study (Purps et al., 2014) analyzed 129 different populations uploaded to the YHRD and found clusters of identical high-resolution haplotypes (23 loci), which are most probably in IBD, for example, in Europe (Finns, Estonians, Roma), in Africa (Maasai, Xhosa) or North America (Athapaskans, Yupik). The majority of sample populations, however, do not show signs of extended cryptic relatedness. The menu "National databases" at the YHRD provides a list of most frequent haplotypes for low (9, 12, 17 markers) and high-resolution Y-STR panels (23, 27, 29 markers) for each of the 135 countries and shows that extended patrilineal relatedness is typical for some regions of the world. To conclude this point: sampling informed by ethnographic knowledge and compliance with the ethical tenets, especially the informed consent, is important to generate trustworthy and usable datasets.

To calculate the Y-STR profile frequency constant and variable estimators can be used. Constant estimators (counts in a database accompanied by a confidence interval) are bound from below by the database size and are, therefore, often overly conservative. In contrast, variable estimators are independent of the database size but not from its composition. The collected samples are used to model the probability distribution of Y-STR haplotypes. Because these distributions differ between metapopulations also the estimated frequency of a certain haplotype differs considerably between MPs (see Figure 2). As mentioned earlier, one calculation method using this principle is the Discrete Laplace (DL) method (Andersen et al., 2013), which is available from the YHRD platform. A similar method, claimed to perform equally well, has been proposed recently (Andersen, Curran, de Zoete, Taylor, & Buckleton, 2018).

Due to its reputation as the most-studied and best-sampled haplotype format ($N = 205{,}059$ in current YHRD release 59), the low-resolution 17-locus YFiler® haplotype (Mulero et al., 2006) is the workhorse and core set of forensic Y testing. The panel is included in all commercial kits, exhibits moderate mutation rates per locus and is, therefore, found in relevant numbers "identical by state" outside families. It is, therefore, advised to use structured databases to estimate YFiler frequencies and calculate match probabilities either using counting or DL. In Figure 5 the observed and DL estimated frequencies of more than 25,000 Western European haplotypes are compared. Obviously the strength of evidence of most YFiler haplotypes becomes magnitudes higher when DL is used instead of counting. Some limitations apply to the DL method. Haplotypes with Null and intermediate alleles cannot be analyzed and multicopy loci are excluded because of the violation of the stepwise mutation model (Kimura & Ohta, 1978) on which the method is based.

In the case report shown in Figure 2 a YFiler haplotype was analyzed using the DL method in different worldwide sampled metapopulations. The probability of occurrence of this haplotype was calculated with DL to be ≈ 1 in 50,000 in Western Europe (a population of ≈ 400 million). According to this numbers roughly 8,000 men unrelated to the suspect could carry an identical haplotype in Western Europe. The same haplotype has a chance to occur in the Chinese Han population with ≈ 1 in 236 million or 5 men in a population of $\approx$1.2 billion. These statements were part of the defense proposition. However, with a panel of additional 30 Y-STRs including rapidly mutation markers the unrelated men wherever they live could be excluded with high credibility and thus only the family members of the suspect remain which were otherwise excluded as the source of the trace DNA.

6 | THE EVIDENTIAL VALUE OF HIGH-RESOLUTION HAPLOTYPES

So-called rapidly mutating Y-STR markers (RM Y-STRs) come with the promise to solve the problem of identical IBD and IBS haplotypes, which is often seen as an obstacle to a wider application of Y testing in forensics (Ballantyne et al., 2012).

TABLE 2 Current metapopulations as defined in the YHRD (release 59)

Metapopulation	Main regions
Eurasian	*Eurasia, North and South America, South Africa, Australia*
Eurasian—European	Europe, North and South America, South Africa, Australia
Eurasian—European—Western European	Western Europe
Eurasian—European—Eastern European	Eastern Europe
Eurasian—European—South-Eastern European	Southeastern Europe
Eurasian—Altaic	Western Asia, Turkey, Central Asia
Eurasian—Indian	India, Pakistan, Bangladesh
Eurasian—Indo-Iranian	Central Asia, Iran, Pakistan, Afghanistan, Europe
Eurasian—Uralic-Yukaghir	Finland, Estonia, Siberia
Eurasian—Caucasian	Caucasus
East Asian	*Eastern Asia*
East Asian—Sino-Tibetan	Eastern Asia
East Asian—Sino-Tibetan—Chinese (Han)	China
East Asian—Sino-Tibetan—Tibeto-Burman	China, Himalaya
East Asian—Korean	Korea
East Asian—Tai-Kadai	Southeastern Asia, Thailand
East Asian—Austronesian	Southeastern Asia, Melanesia
East Asian—Japanese	Japan
East Asian—Austro-Asiatic	Southeastern Asia
East Asian—Dravidian	Southern India, Sri Lanka
East Asian—Indo-Pacific	Polynesia
Afro-Asiatic	*Northern Africa, Near East, Eastern Africa*
Afro-Asiatic—Semitic	Northern Africa, Near East
Afro-Asiatic—Cushitic	Somalia, Ethiopia, Eritrea
Afro-Asiatic—Berber	Northern Africa
African	*Africa, Northern and Southern America*
African—Sub-Saharan	Subsaharan Africa
African—Afro-American	Northern America
African—Afro-Caribbean	Caribbean America
Native American	*Northern and Southern America*
Australian Aboriginal	*Australia*
Eskimo Aleut	*Northern America, Greenland*
Admixed	*None*

For rapidly mutating markers like DYS518, DYS627 DYS570, and DYS576 mutation rates are measured in the range of 1.14–1.46 x 10^{-2}, which are about 10 times higher than rates for common markers like DYS19, DYS389I, DYS390, and DYS391, which are in the range of 2.08–2.72 x 10^{-3} (Source https://yhrd.org/pages/resources/mutation_rates). The elevated mutation rates make haplotypes consisting of RM Y-STRs strikingly more discriminative in comparison to low-resolution haplotypes and have the potential to differentiate even close paternal relatives (Ballantyne et al., 2012, 2014). Pure RM Y-STR multiplex assays have been introduced (Alghafri et al., 2015) and current commercial kits include RM markers as well (Gopinath et al., 2016; Thompson et al., 2013). However, as shown in the case report mentioned earlier even distantly related family members cannot be individualized securely with high-resolution haplotypes. Rather, the higher mutation rates of modern kits reduce the proportion of coincidentally matching IBS haplotypes of random men close to zero and match only the patrilineal relatives. Based on results from simulations recent publications suggest, that the probability of finding a matching

high-resolution haplotype rests almost completely with a relative (Andersen & Balding, 2017; Taylor, Curran, & Buckleton, 2018). The number of male relatives which carry an identical high-resolution haplotype is calculated (by simulation) to be less than 40 (Andersen & Balding, 2017). According to these authors this number is irrespective of a frequency retrieved from a database. In the example case above, a statement on the evidential value of the 47-locus haplotype match between the trace and the suspect would then read as follows: "*The suspect or someone from his family is the source of the trace. Persons unrelated to the family can be excluded as the source of the trace.*"

It remains to be seen whether the courts will follow this argumentation. We should remember that Y-STR testing passed admissibility standards, for example, in the USA because its analysis and statistical treatment is not fundamentally different and analogous to autosomal STRs. Typing, nomenclature, mutation mechanisms - all this is not different from autosomes. Even the dependency structure of Y-STRs is not a striking difference, the Y chromosome as a whole is one forensic marker (though extremely variable) among others. The shared identity by descent makes the interpretation of haplotypes difficult, but—as in the autosomal case—more markers, good databases and advanced frequency estimation methods can provide reliable and fair results.

7 | ANCESTRY INFERENCE USING THE Y CHROMOSOME

The ability to predict male ancestry in a forensic context relies on the beauty and logic of the universal phylogenetic tree of Y chromosomes. All sequences typed for unique event polymorphisms (usually SNPs) can be linked to this tree. Recently an extra-large phylogeny incorporating 13,261 high-confidence SNPs retrieved from 3.7 Mb of the male-specific part of the Y chromosome sequenced in 448 human males has been published (Hallast et al., 2015). However, for forensic geneticists a reduced consensus tree is a more pragmatic choice (Larmuseau, Van Geystelen, Kayser, van Oven, & Decorte, 2015). Such a tree has a clear structure and nomenclature and captures the main Y-haplogroups of worldwide significance (van Oven, Van Geystelen, Kayser, Decorte, & Larmuseau, 2014) and is online available at http://phylotree.org/Y/ or via https://yhrd.org/pages/resources/ysnps). Using simple methods as minisequencing the ancestry of an unknown trace donor can be analyzed using a small number of SNPs typed in a hierarchical manner (Geppert & Roewer, 2012). In a first round SNPs for the early diverging branches are used to exclude all strata except one, followed by a second multiplex PCR with SNPs designated to resolve the branch further. The most probable place and time of the expansion spreading the detected regional founder mutation is than analyzed using reference data and dating methods (Wei, Ayub, Xue, & Tyler-Smith, 2013). In a third round, Y-STRs and YHRD can be applied to resolve the Y chromosome down to the familial (genealogical) level. Provided that representative reference data exist for the located part of the world, the region and ethnic group of the paternal ancestors can be ascertained reliably. For example, the basic haplogroup C defined by the SNP M130 is widespread in Central and East Asia, Australia and the Americas. Mutations further down the C branch arose later and are, therefore, restricted to lately occupied parts of the New World, for example, the derived American founder mutation C3 defined by SNP MPB373 occurs only in

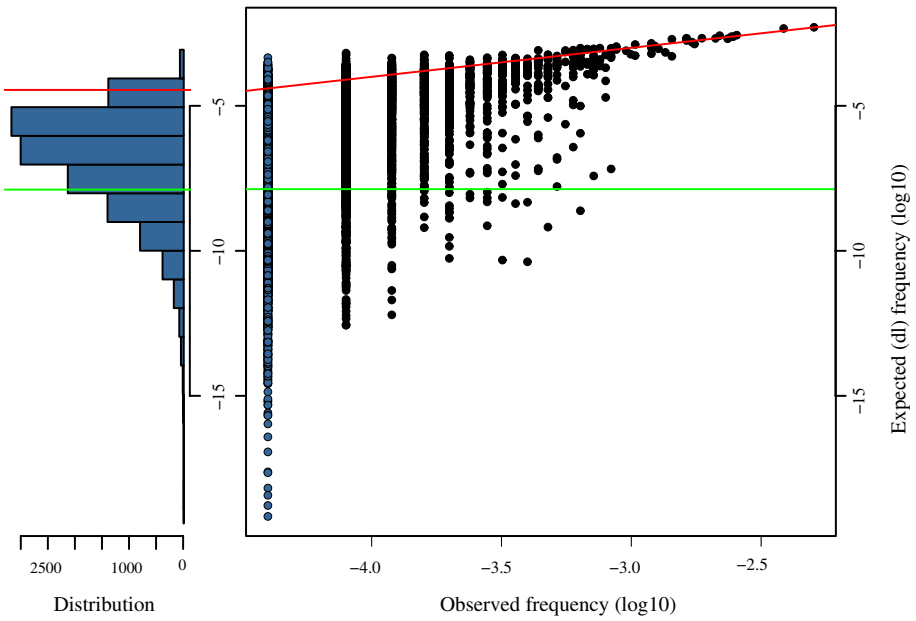


FIGURE 5 Comparison of observed and DL-estimated frequencies of all 15-loci haplotypes in the Western European metapopulation (YHRD, release 55, $N = 25{,}665$ YFiler haplotypes excluding the multicopy locus DYS385ab). The red line depicts concordance between observed and estimated haplotype frequencies; the green line marks the threshold of the DL frequency estimation for very rare haplotypes in this MP which is based on the distribution of singletons frequency estimates. To the left, the density distribution of the singleton estimates is plotted (modified from Roewer & Willuweit, 2018)

native groups living in the Northwest of South America (Pinotti et al., 2018). In Europe, collaborative exercises with blind samples are annually organized since 2017 in order to validate ancestry prediction methods for forensic use. The participants from university institutes and police used most frequently Y chromosomal SNPs and STRs, mitochondrial control region sequences and autosomal ancestry informative markers (AIMs) (manuscript in preparation).

8 | INVESTIGATIVE LEADS PROVIDED BY Y-CHROMOSOME TESTING

As outlined earlier in this text Y chromosome analysis can detect hidden male DNA on a strong background of female DNA. A Y-STR screening is, therefore, advisable in all cases where a male DNA left by a perpetrator is expected but not found in regular autosomal STR screening. Samples with full Y could then be sensitively reanalyzed for autosomal markers in order to develop a profile eligible to search in a national DNA database. Most databases do not upload Y-STR profiles because of the ambiguity of a match, which would hit also a group of paternal relatives. However, some countries, for example, China (Ge et al., 2013, 2014) started to accumulate also Y-STR haplotypes in their national DNA databases. In severe crime cases Y chromosome analysis can be seen as a "last resort" technique to collect intelligence information on an unknown perpetrator and to narrow down the number of suspects (Qian et al., 2017; Yao et al., 2016). In Europe a Y-STR profile was used in the elimination process of a mass screening already in 2000 (Dettlaff-Kakol & Pawlowski, 2002) when a serial rapist and murderer called the "Vampire from *Świnoujście" was* hunted down in Northern Poland. About 11,000 men were screened and one man was found who had an identical profile in all Y-STR loci analyzed for the trace. He was, however, excluded based on autosomal mismatches but the identity of the Y profile suggests that the real rapist and the typed volunteer were closely related males. Analysis of the reference DNA obtained from the man's brother revealed an identical Y-STR and autosomal profile to that identified at the crime scenes.

The combined strategy of Y-based ancestry prediction and familial searching was applied in another rape and murder case from 1999 in the Netherlands (Kayser, 2017). By comparing the Y-STR profile left at the scene with samples in the YHRD and other references it was concluded that the trace donor's paternal ancestors are likely to originate from the Northwestern part of Europe. The immediate use of such information was not legal in the Netherlands until 2003 when a new law regulates the forensic use of DNA information regarding bio-geographic ancestry for investigative purposes. Finally, in 2012 a large dragnet was organized and 7,600 men living in close vicinity to the crime scene were invited to provide a cheek sample. The police decided to apply Y-STR profiling in this dragnet. Assuming that the perpetrator would not participate, the Y-STR analysis could identify paternal relatives who share the same Y-STR profile. In this case, however, the perpetrator, a local man, was among the volunteers evidenced by both by a Y-STR and an autosomal STR match.

In several countries, the Y chromosome in combination with other parts of the genome is used for the identification of unknown corpses or dismembered body parts of missing persons (Ambers, Votrubova, Vanek, Sajantila, & Budowle, 2018). In such cases DNA-based intelligence often competes with non-DNA methods like isotope analysis or facial reconstruction. The disparities between different forensic methods in regard to the scientific basis as highlighted in the NRC report (NRC, 2009) is obvious, but rather to forensic scientists than to the law enforcement. It is, therefore, very important to continue to strengthen the science behind modern DNA analysis methods including Y chromosome testing and to propagate its achievements.

9 | SUMMARY

Today, Y chromosome markers are used to assist forensic investigations in a broad range of casework, either for rapid sexing of traces, for the identification and profiling of masked male DNA component(s) in mixtures with an excess of female DNA, for tracing male lineages in relationship testing or for analyzing the deep-rooting ancestry of unidentified corpses or unknown trace donors. Like any forensic DNA analysis, judging the significance of a match depends upon the population frequency of the profile. As has been discussed in this review, Y chromosomes are highly geographically structured and there is the further complication that all members of a patrilineage are expected to share Y haplotypes. The forensic community has been proactive in establishing large quality-assured and publicly accessible databases of Y-haplotypes to address these issues. The international cooperation in the YHRD project is a blueprint for collaboration and community building in the field of forensic genetics.

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CONFLICT OF INTEREST

The author has declared no conflicts of interest for this article.

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